

Manual

Personnel Scheduling Reports PEP-APP 8.1

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1 Personnel Scheduling Reports - Overview

Purpose

This function package contains personnel scheduling evaluation and planning functions.

Implementation Considerations

Use this function package to:

- compare the personnel requirement with the available personnel capacity in graphic planning;
- schedule the assignment of employees to workplaces in HYDRA and display the resulting personnel schedule.

Integration

Use of this function package requires function package *Personnel Scheduling Administration Functions* (PEP-VWF).

Features

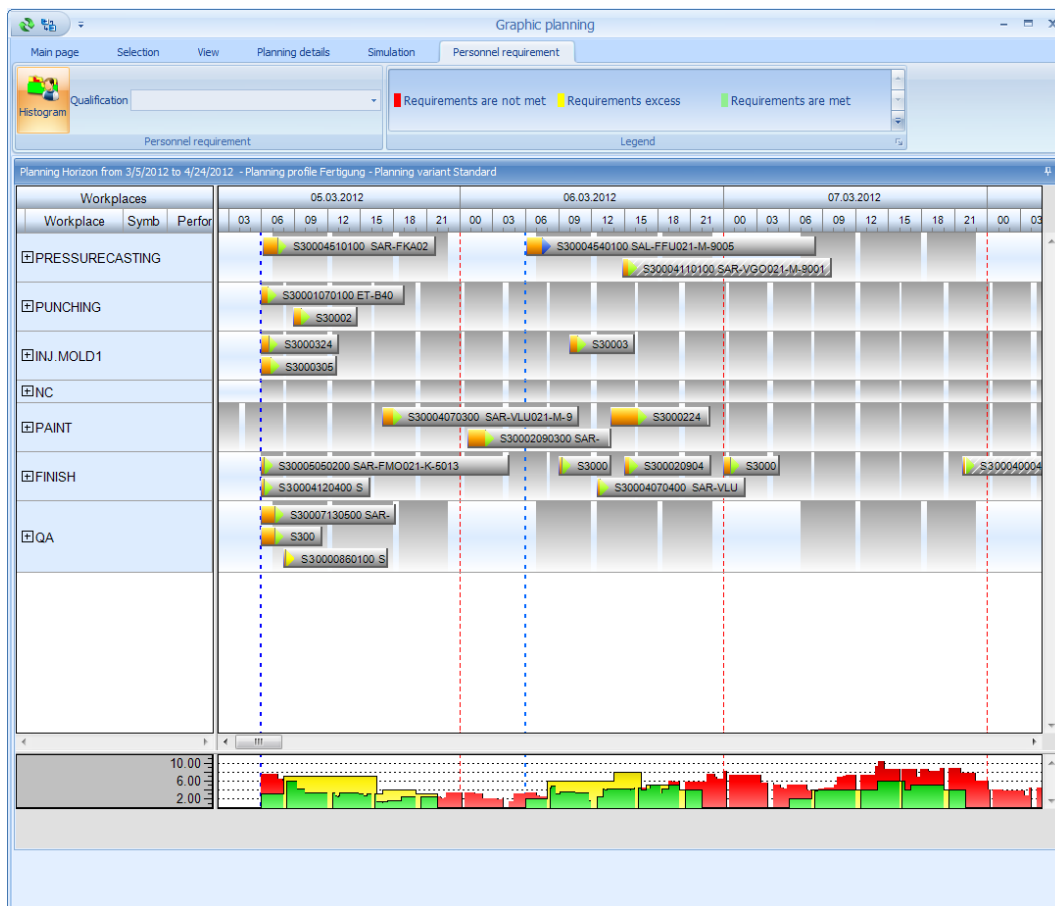
- Overview of workforce requirements in graphic detailed scheduling
 - Graphical display of shift related personnel requirement with variable time horizon and comparison of available people with their qualifications
- Staffing
 - Graphical display of workplaces with personnel requirement and available personnel capacity
 - Shift related assignment of employees to workplaces
- Personnel schedule
 - Personnel schedule in tabular format with a list of assigned workplaces

2 Graphic Planning - Workforce Requirements

Overview

Menu	Production Control → Production Preparation → Graphic Planning
Index tab	Personnel Requirement
Function authorization	perov

The function "Personnel Requirement" can be used to display the personnel requirement as a histogram within the graphic planning and to compare this to the planned available personnel.



Depending on whether the personnel requirement can be met or whether too many people have been scheduled, the resulting areas are displayed in different colors:

Color	Meaning
	The personnel requirement is not met
	Too many people scheduled
	The personnel requirement is met

The chart helps the user to schedule orders in periods with sufficient available personnel capacity.



Should more than one OP be scheduled for a workplace (HLS), the personnel requirement for these OPs is summed.

Prerequisite

A prerequisite to display the planned personnel capacity is that the absence and shift planning data has been maintained correctly in HYDRA. This is usually done through the [Labor time maintenance](#).

Additionally, the personnel requirement must be defined. This can be performed using one of 3 procedures:

- [Workforce requirements of workplaces](#)
- Personnel requirement in the Machine/operator relation (M/O relation setup, M/O relation production) of the [Operations](#)
- Personnel requirement in the [Production resources and tools](#) (Resource types) *PRU* for setup and *PER* for completion)

Tool bar

The index tab *Personnel Requirement* contains the following functions:



Display Personnel Requirement

This button is used to switch the personnel requirement histogram on and off. The orders must be reloaded after switching the histogram on to obtain the orders' personnel requirement.

Qualification

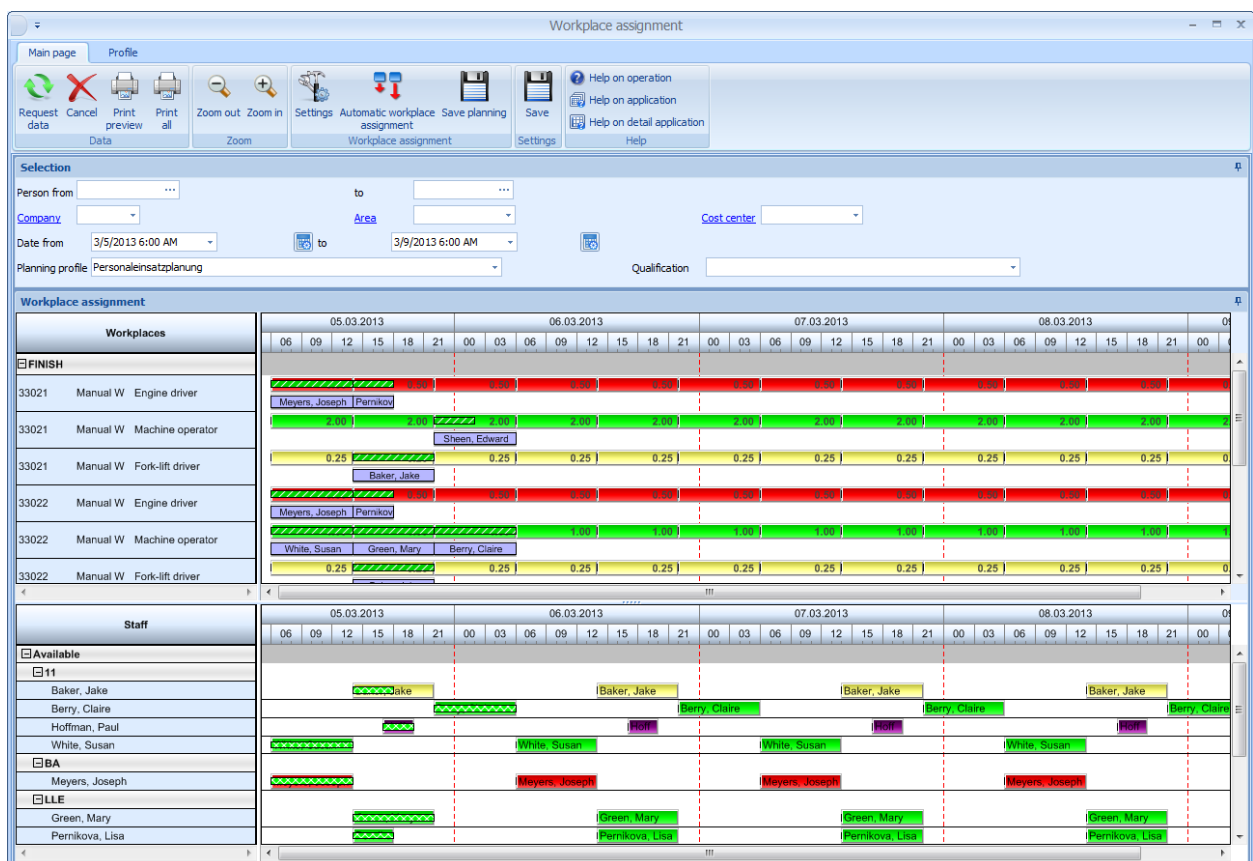
The personnel requirement display can be restricted to one or more qualifications. Displaying all qualifications shows whether enough personnel is available. Restricting the display to a qualification shows whether enough personnel with this qualification is available.

3 Workplace Assignment

Summary

Menu	Production Control → Workforce Requirements Planning → Workplace Assignment Human Resources Management → Workforce Requirements Planning → Workplace Assignment
Transaction code	wpas
Function authorization	wpas

Staff can be planned on workplaces using the workplace assignment function. Planning can be performed manually as well as automatically.



Display

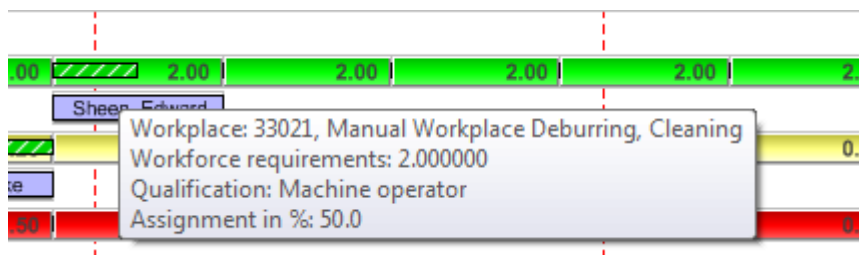
The center of the application shows the workplaces including the required qualifications grouped by workplace groups. The workforce requirements for the relevant qualification are shown for each shift within the bar. The bar length depends on the planned year model or the year model of the [workplaces](#). Only the workplaces requiring personnel are displayed.



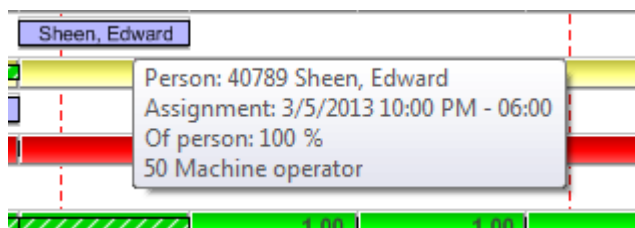
The breaks resulting from the employees' planned working time and that of workplaces are not considered in "workplace assignment". Workplace assignment plans whole shifts instead to prevent the bars for employees and workplaces from being interrupted.

The lower section of the workplace assignment shows the available staff grouped by areas and the unavailable staff grouped by the abbreviations for absence reasons. The bar length depends on normal working times (beginning of normal time and end of normal time) from the planned working time day type.

A green bar (hatched) shows the assignment rate in percent per shift at a workplace. The shift tooltip shows the percentage rate as figure. In addition, the planned staff is displayed:



The tooltip of an assigned person provides information about the person, their qualifications and assignment:



There are different ways of presentation for the assigned staff:

Presentation	Description
	The person provides the required qualification
	The person does not provide the required qualification
	The assignment exceeds the workforce requirements of the workplace

Prerequisites

Absence time planning and shift planning need to be maintained properly in HYDRA to be able to show the planned personnel capacities. As a general rule, this is done in [labor time maintenance](#).

In addition, the workforce requirements need to be defined. There are three possibilities:

- [Workforce requirements of workplaces](#)

- Workforce requirements of the machine/operator relation (M/O relation for setup, M/O relation for production) of [operations](#)
- Workforce requirements of [production resources and tools](#) (resource types *PRU* for setup and *PER* for production)

Operations have to be planned in detail if the planned operations are to be taken into account by workplace assignment. This can be performed, for example, in graphic planning. As an alternative, the ERP system might already have planned the operations in detail to send them to HYDRA.

Selection criteria

The application provides the following selection criteria:

Qualification

The workforce requirements to be displayed can be restricted to specific qualifications.

Planning profile

Using the planning profile, personnel assignment can be restricted to specific workplace groups and employees.



This field is only available if the additional function "enhanced selection and visualization" (PEP-ESV) is available.

Toolbar



Zoom in,

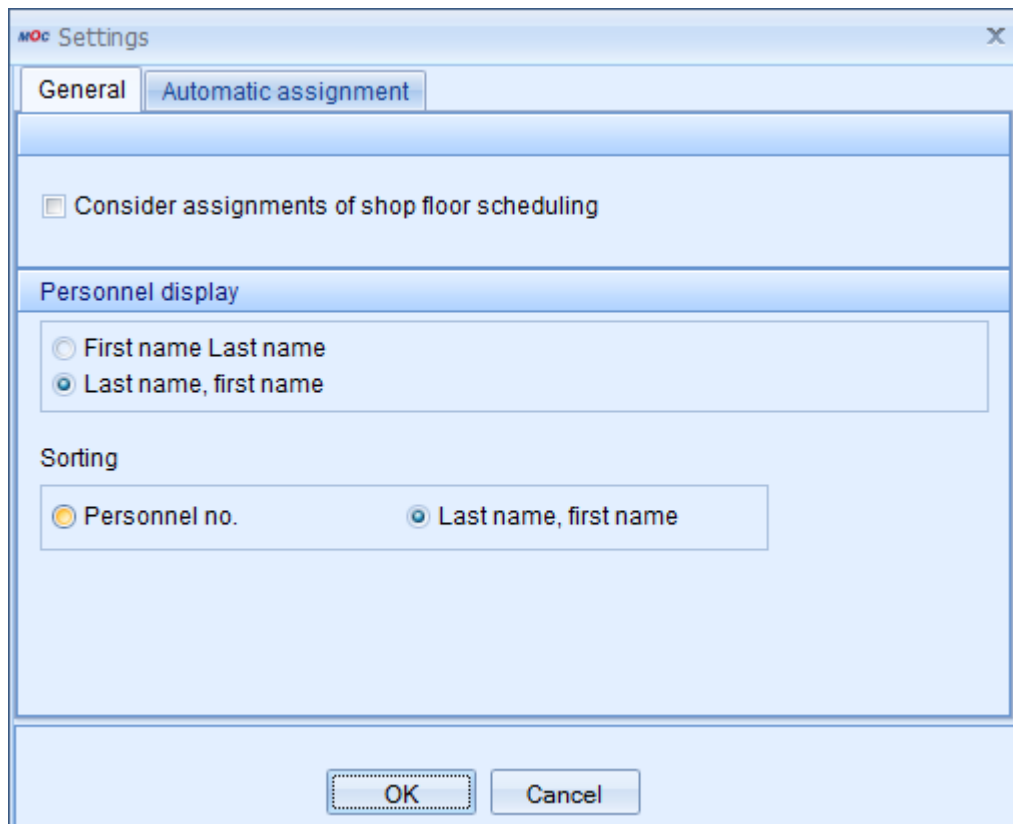


Zoom out

These buttons reduce or increase the displayed workplaces and persons.



Settings



The screenshot shows a 'Settings' dialog box with a title bar containing a close button (X). Inside the dialog, there are two tabs: 'General' and 'Automatic assignment'. The 'Automatic assignment' tab is currently selected. Below the tabs, there is a checkbox labeled 'Consider assignments of shop floor scheduling' which is currently unchecked. Below this, there is a section titled 'Personnel display' containing two radio button options: 'First name Last name' and 'Last name, first name'. The 'Last name, first name' option is selected. Below this section is a 'Sorting' section with two radio button options: 'Personnel no.' and 'Last name, first name'. The 'Last name, first name' option is selected. At the bottom of the dialog, there are two buttons: 'OK' and 'Cancel'.

Consider assignments of shop floor scheduling

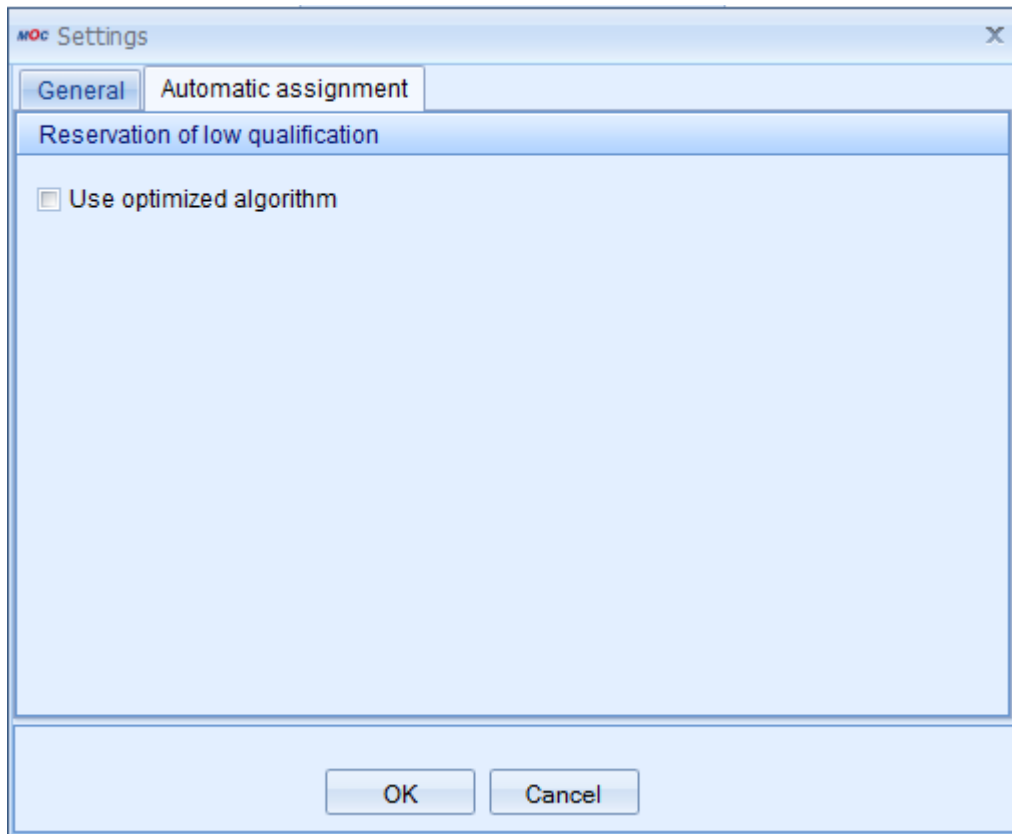
If this option is enabled only those shifts for which an operation is planned are shown for the workplaces. If it is not the entire shift that is assigned to operations, the workforce requirements of the shift will be reduced proportionately.

First name last name or last name, first name

This field defines whether the last name is shown in front of or behind the first name.

Sorting

This option specifies whether the staff grouped by areas is sorted by the personnel number or by the name within the relevant area.



Use optimized algorithm

This option enables the [optimized, automatic assignment](#).



The configurations are saved with respect to the user per workplace.



Automatic workplace assignment

Calls up the [automatic workplace assignment](#)

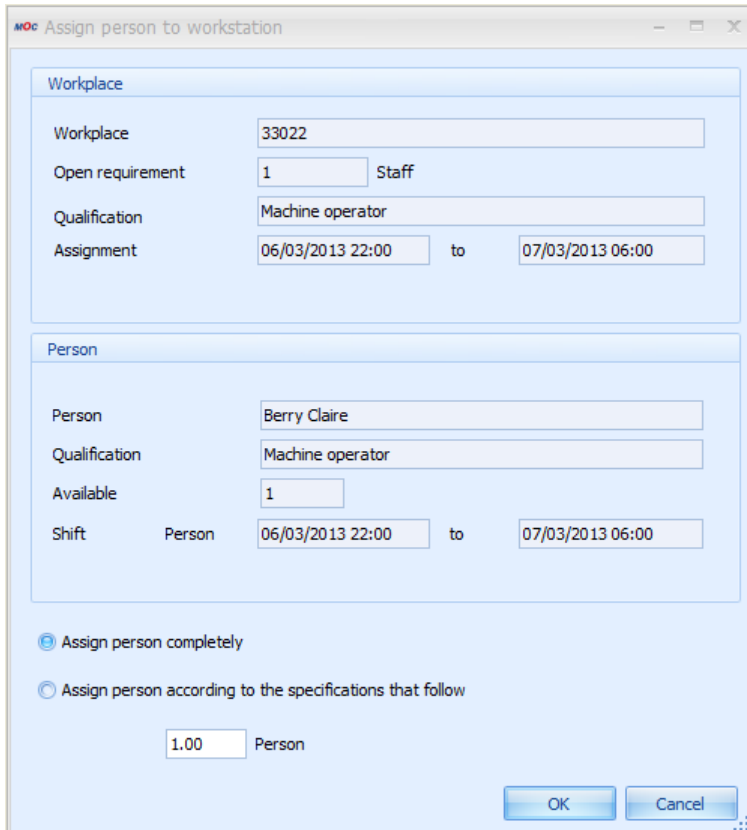


Save planning

This function saves the current workplace assignment, which is then available to all users.

Manual assignment

Staff can be planned manually by way of "drag & drop". To do so, select the bar of a person, hold the mouse button down and drag it to the top to a shift of a workplace. The below dialog appears, once the mouse button has been released:



Workplace

Workplace: 33022

Open requirement: 1 Staff

Qualification: Machine operator

Assignment: 06/03/2013 22:00 to 07/03/2013 06:00

Person

Person: Berry Claire

Qualification: Machine operator

Available: 1

Shift: 06/03/2013 22:00 to 07/03/2013 06:00

☒ Assign person completely

☐ Assign person according to the specifications that follow

1.00 Person

OK Cancel

Workplace

Shows the workplace including the required qualification and the open requirements.

Person

Shows the person with the qualification and availability.

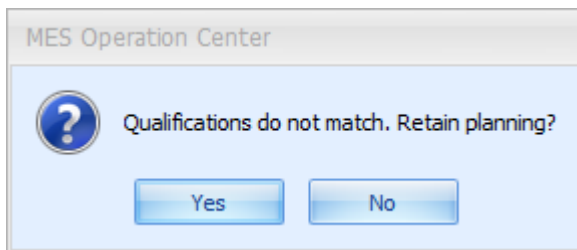
Assign person completely

The person's available time is completely assigned to the workplace, irrespective of whether this exceeds the workplace requirements or not.

Assign person according to the specifications that follow

The missing assignment is compared with the person's availability and the result is shown in this field. The suggested value may be changed.

If the person does not have the required qualification, this message appears:



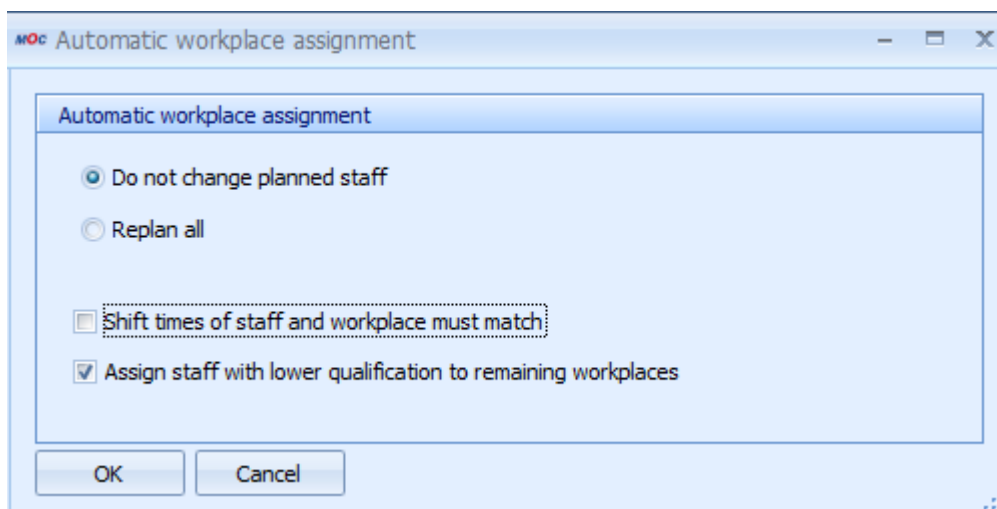
It is now up to the user to decide whether or not to perform this assignment.



To remove a person from planning, the personnel bar only has to be selected and dragged back from the workplace to the pool of staff.

Automatic assignment

The below dialog appears when starting automatic workplace assignment:



Do not change planned staff, replan all

These options specify whether already existing assignments are to be kept by automatic assignment or whether they are to be removed from planning to be planned once more.

Shift times of staff and workplace must match

Staff is only planned on workplaces with shift times matching the employee's planned normal working time.

Assign staff with lower qualification to remaining workplaces

All staff qualifications are taken into account as part of the assignment. Assignment is performed sorted by the order of priority of the employee's qualifications. If this option is not enabled, only qualifications of the highest order of priority (99) are assigned.

Optimized, automatic assignment

The optimized, automatic assignment function can be enabled by the [#settings](#).

In contrast to the automatic assignment function described above that only considers the order of priority for qualifications, the optimized, automatic assignment function also considers the person's flexibility. This is to achieve a higher assignment rate for automatic planning. The optimized assignment function works as follows:

Plan staff with only one assignment option

At first it is checked whether there is staff with only one possible assignment option. This assignment is performed, irrespective of whether there are employees having a higher order of priority for the required qualification. Consequently, this person (with only one assignment option) does not remain unassigned.

Assignment of workplaces with only one possible operator

Then it is checked whether a qualification required by a workplace can only be met by one employee. This employee is assigned to the workplace, irrespective of whether this person still has additional qualifications of a higher order of priority. Consequently, the qualification required by this workplace does not remain unassigned.

Assignment due to the order of priority of qualifications

Just as it is the case with the "normal" automatic assignment function, now the employees are assigned to the workplaces sorted by the order of priority of their qualifications. However, every time an assignment is made, the two previous steps are taken again. For by assigning an employee, there might again be persons with only one possible assignment option or workplaces with only one possible operator.



The optimized algorithm is only used if the option "Assign staff with lower qualification to remaining workplaces" is enabled for automatic assignment.

4 Workforce Requirements Plan

Summary

Menu	Human Resource Management → Personnel Scheduling → Personnel Schedule
Transaction code	pesch
Function authorization	pesch

The personnel schedule display shows which person is assigned to which workplace and if there are still any unassigned, free times among the employees.

Workforce requirements plan

Main page Profile

Request data Cancel Print preview Print all Save Help on operation Help on application Help on detail application Help

Selection

Person from ... to ...

Company ... Area ... Cost center ...

Date from 3/5/2013 to ...

Planning profile ...

Workplace ... Summarize absences Display days without normal time

Workforce requirements plan

Drag a column here to group by that column

Person	Name	Shift type	Absence	Weekday	Date	Time	End of work...	Workplace	Designation
667	Smith, Christian		FLX	Tuesday	3/5/2013	8:00	16:30		
40256	Pernikova, Lisa	S	HLD	Tuesday	3/5/2013	14:00	22:00		
40563	Green, Mary	S		Tuesday	3/5/2013	14:00	22:00	33021	Manual Workplace Debur...
40789	Sheen, Edward	N		Tuesday	3/5/2013	22:00	6:00	33021	Manual Workplace Debur...
50014	Albert, Claudia			Tuesday	3/5/2013	8:00	14:00	33021	Manual Workplace Debur...
50014	Albert, Claudia			Tuesday	3/5/2013	14:00	16:30	33021	Manual Workplace Debur...
50201	Mayer, Hugo Egon	N		Tuesday	3/5/2013	22:00	6:00		
59881	Brown, Albert			Tuesday	3/5/2013	8:00	16:30		
85741	Meyers, Joseph	F		Tuesday	3/5/2013	6:00	14:00	33021	Manual Workplace Debur...
85741	Meyers, Joseph	F		Tuesday	3/5/2013	6:00	14:00	33022	Manual Workplace Debur...
93856	Miller, Steven			Tuesday	3/5/2013	8:00	16:30		
96665	Hanson, Betty			Tuesday	3/5/2013	8:00	14:00	52276A	Paint Line 1a
96665	Hanson, Betty			Tuesday	3/5/2013	14:00	16:30	52276A	Paint Line 1a
10002541	Berry, Claire	N		Tuesday	3/5/2013	22:00	6:00	33022	Manual Workplace Debur...

Selection criteria

The following selection criteria are available in the application:

Person from/ to, company, area, cost center

Selection of the people to be displayed

Date from/ to

Period for which the personnel schedule is to be displayed.

Workplace

Machine/ workplace

Planning profile

The display of the personnel schedule can be limited using a planning profile. This field can only be accessed if the additional function Advanced selection and visualization is present.

Summarize absences

Contiguous days with absences for a person are output in a row with the start and end date.

Display days without normal time

The list also includes days off on which a normal time is not stored in the working time day type of HYDRA-PZE. In this way, in a longer evaluation period one calendar view per person is possible.

If the personnel schedule was limited using a planning profile, only those days are displayed on which the employees have an assignment for a workplace of the corresponding planning profile.



The value in the *Free time* column contains only the assignments within this planning profile and for this reason it is only correct if the employee was not assigned outside of the planning profile.